# Transparent Multi-Hop Routing for Raft

## Under Partial Network Partitions

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# Partial partitions can interrupt Raft

## Nodes may remain reachable through peers even when their direct links fail.

Consensus services need progress to remain available.

### RQ1 Progress

When does forwarding restore stable commits?

### RQ2 Latency

What delay does forwarding add to client-observed commits?

Goal: improve resilience while keeping the Raft protocol unchanged.

# Raft and the routing gap

Raft commits after a majority stores an entry.
A five-node cluster needs three replicas.

| | |
|---|---|
| **Jensen et al. (2021)** | Propose overlays for Raft |
| **Nifty (2020)** | Reroutes traffic in other systems |
| **Omni-Paxos (2023)** | Changes consensus for partial connectivity |

**Our gap: measure routing benefits and costs beneath unchanged Raft.**

# Controlled network experiments

**900**

validated trials

5 topologies
2 communication modes
3 link delays
30 repetitions

## HashiCorp Raft in Mininet

Direct links vs. preconfigured forwarding

Link delay: 0.5, 5 or 20 ms
Heartbeat timeout: 500 ms
PreVote disabled

100 requests/s, 64-byte payloads

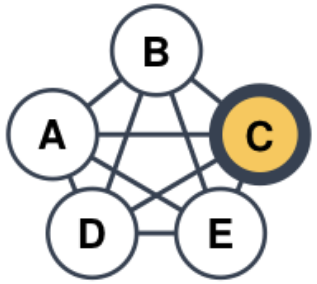
30 s warm-up, cut links, observe 30 s
T3: reseed and restart after injection

All 900 trials validated across the complete delay matrix.

# Network configurations

## T0

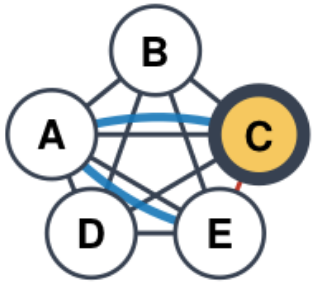
### Healthy network




All direct links work

## T1

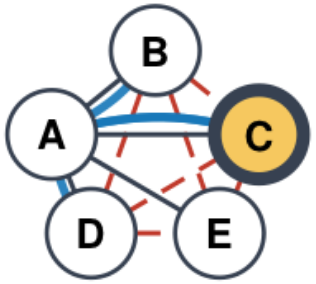
### Non-critical cut




Direct majority remains

## T2

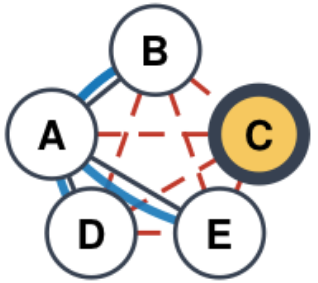
### Quorum loss




Direct majority is lost

## T3

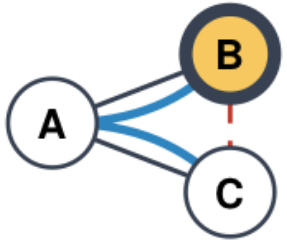
### Constrained election




A has an older log

## T4

### Three-node chain



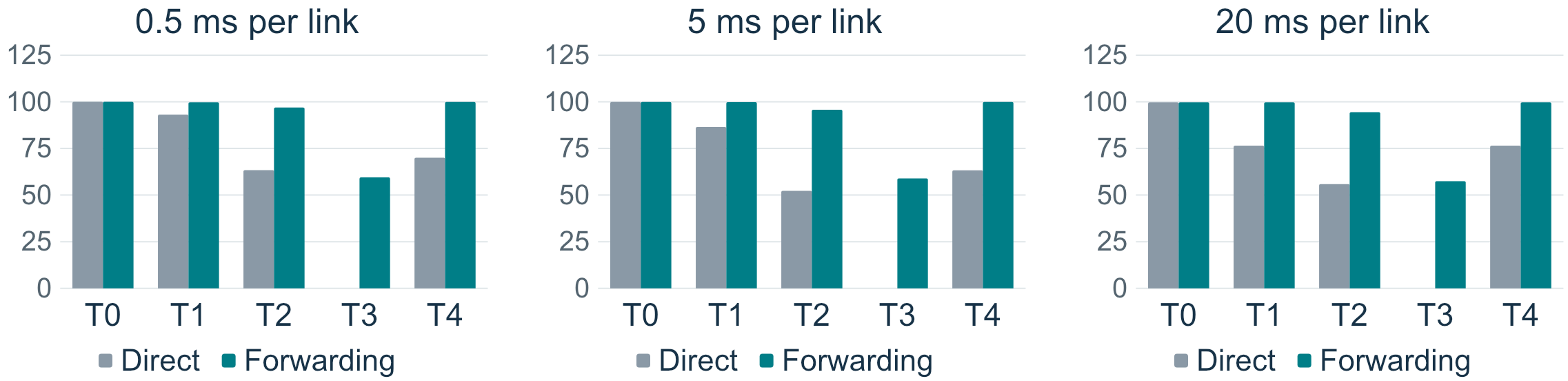

B and C connect through A

Surviving link  Failed link  Forwarding path  Initial leader

# Forwarding sustains client progress

## Stable-progress heuristic: forwarding 450/450, direct 309/450

Mean successful client commits/s



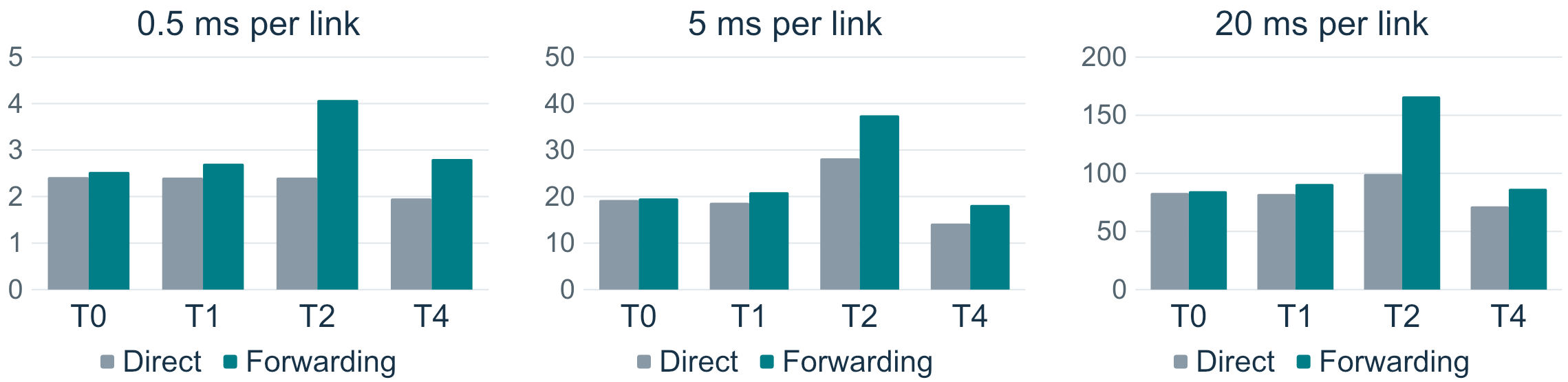

**T2 throughput improves by 53%, 83% and 69% across the delays.**

T3: all 90 forwarding trials commit. Direct mode commits in none.

n = 30 per condition. T3 throughput includes restart and setup time.

# A forwarded quorum adds commit latency

Mean of per-run median successful-request latency (ms)




**T2 direct / forwarding: 2.41 / 4.08, 28.23 / 37.49, 99.47 / 166.21 ms**

T3 forwarding: 3.57, 35.81, 147.77 ms. No direct latency samples.

Axes use different scales. T1/T4 direct summaries exclude trials with no commits.

# Routing can improve resilience at a cost

**Progress**

## Forwarding restores T3 commits and improves T2 throughput.

**Cost**

## Commits take longer when the majority needs extra hops.

T1/T4 gains also reflect client access to leaders that retain quorums.
T3 recovery includes setup. Results cover one workload and timeout.

# Future work

### 01 Client reachability and recovery timing

Control client placement and separate setup from Raft recovery.

### 02 Heartbeat timeout and path delay

Vary the timeout to measure its interaction with network delay.

### 03 Deployment across machines and regions

Test variable delays in a more realistic network.

# Questions

Transparent routing for Raft under partial partitions